Zishuo Zhao

LLM Agents | Reinforcement Learning | NLP | Multimodal Learning

Alibaba International | Hangzhou, China
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Research Interests

Large language model agents and reinforcement learning, with an emphasis on multi-turn preference learning, turn-level alignment, planning, reasoning, and tool use. Additional interests include source-free domain adaptation and multimodal retrieval.

Experience

Alibaba International

LLM Agents & Reinforcement Learning | Jul 2024 - Present

- **Customer service agents:** Contributed to AliExpress pre-sales and after-sales agents, including planner-reasoner architectures, tool calling, reward evaluation, and preference alignment using DPO and GRPO.
- **Multimodal review:** Contributed to image-based review workflows using multimodal models, OCR, image classification, and preference alignment.
- **Voice agents:** Worked on Lazada inbound customer service and ICBU outbound voice agents, including planning, knowledge grounding, interruption handling, and order understanding.
- **Agent learning:** First-author work on Turn-level Multiscale Density Ratio Estimation for LLM Agents (arXiv:2609.16760).

Alibaba International

Research Intern | Dec 2023 - Feb 2024

Explored RLHF and DPO for customer service language models, reinforcement learning frameworks, and automated model evaluation.

Tencent AI Lab

Research Intern | May 2023 - Oct 2023

Worked on LLM role-playing, real-time game commentary, and report generation. Participated in research on PPO-based methods for extending model input context.

Education

Sun Yat-sen University

Master's degree in Intelligent Control | Sep 2021 - Jun 2024

School of Intelligent Systems Engineering | Shenzhen, China

Sun Yat-sen University

Bachelor's degree in Computational Materials | Sep 2017 - Jun 2021

School of Materials | Guangzhou, China

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Publications & Research

Preprints

Turn-level Multiscale Density Ratio Estimation for LLM Agents

Zishuo Zhao, Kai Chen, Ao Li, Yuan Liu

arXiv:2609.16760, 2026

FLEX-CLIP: Feature-Level GEneration Network Enhanced CLIP for X-shot Cross-modal Retrieval

Jingyou Xie, Jiayi Kuang, Zhenzhou Lin, Jiarui Ouyang, **Zishuo Zhao**, Ying Shen

arXiv:2411.17454, 2024

Conference Papers

Source-free Domain Adaptation for Aspect-based Sentiment Analysis

Zishuo Zhao, Ziyang Ma, Zhenzhou Lin, Jingyou Xie, Yinghui Li, Ying Shen

LREC-COLING 2024, pp. 15076-15086

Source-Free Unsupervised Domain Adaptation for Question Answering

Zishuo Zhao, Yuexiang Xie, Jingyou Xie, Zhenzhou Lin, Yaliang Li, Ying Shen

ICASSP 2023, pp. 1-5

Hierarchical Type Enhanced Negative Sampling for Knowledge Graph Embedding

Zhenzhou Lin, **Zishuo Zhao**, Jingyou Xie, Ying Shen

SIGIR 2023, pp. 2047-2051

Multimodal Graph Learning for Cross-Modal Retrieval

Jingyou Xie, **Zishuo Zhao**, Zhenzhou Lin, Ying Shen

SDM 2023, pp. 145-153

Skills

Programming & frameworks: Python, verl
Methods: DPO, GRPO, RLHF, LLM agent development, multimodal learning
Languages: English (CET-6, CET-4)